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Psychological stress and pancreatic cancer patients: a qualitative systematic review protocol

Ann M. Mazzella Ebstein[1,2], **Simi Jesto Joseph**[2,3], **Marisol Hernandez**[1]

[1]Memorial Sloan Kettering Cancer Center, New York, USA

[2]The Center for Translational Research: a Joanna Briggs Institute Centre of Excellence

[3]GI Solutions, Inc, Chicago, USA

Abstract

Objective: The primary objective of this review is to analyze and synthesize the best available evidence on the experiences and perceptions of psychological stress reported by pancreatic cancer patients at any time point from pre-diagnosis, diagnosis, treatment, post-treatment and/or follow-up care.

Introduction: A cancer diagnosis is known to be life-threatening, altering and limiting, and negatively affects an individual's activities of daily living. Despite developments in treatment options for pancreatic cancer patients, it represents the highest mortality and morbidity among cancers. Stress is a subjective phenomenon that negatively impacts an individual's psychological and emotional well-being, and interferes with the ability to cope with cancer symptoms and treatments. Identifying a patient's experience of stress could facilitate educational, spiritual and social resources to address his or her emotional and psychological needs.

Inclusion criteria: Qualitative studies that include individuals with pancreatic cancers, regardless of age, sex or ethnicity, will be considered for inclusion in this review.

Methods: The databases to be searched include PubMed, CINAHL, Cochrane, Web of Science, Embase, Scopus, BioMed Central and PsycINFO. The search for gray literature will include Biosis, OpenGrey, Open Access Theses and Dissertations, and WorldCat. This systematic review will consider all published and unpublished studies with no date limitations. Selected studies will be assessed for methodological quality by two independent reviewers. Coding will be assigned to synthesize any differences in the experiences and perceptions of psychological stress at four time points. Where textual pooling is not possible, conclusions will be presented in narrative form.

Keywords

Emotional distress; emotional stress; pancreatic cancer; pancreatic neoplasm; psychological stress

Correspondence: Ann M. Mazzella Ebstein, mazzella@mskcc.org.

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Introduction

Pancreatic cancer has high mortality and morbidity rates among cancer-related diagnoses. Pancreatic cancers are associated with poor prognosis and the highest levels of stress.[1,2] It is the fourth most common cancer in North America, and has the lowest survival rate.[2] Although progress has been made in treatment of pancreatic cancer, the prognosis for 5-year and 10-year survival rates is approximately 19% and 10%, respectively.[3] Cancerous tumors, such as pancreatic cancers, can also affect an individual's physiology extensively, due to stress responses.[4] The diagnosis and treatments associated with pancreatic and other cancers can lead to significant stress for patients, as they are associated with adaptive behaviors, which may manifest in endocrine, immune, cognitive, emotional and physiological systems.[4,5] These symptoms may impact various stages of the disease process including predictive factors, undiagnosed pathology markers, and clinical events in other diseases.[5]

A cancer diagnosis is known to be life-altering and limiting,[6] and negatively affects an individual's activities of daily living. How an individual reacts and adapts to the initial cancer diagnosis may be influenced by pre-existing psychological factors, such as social and psychological support and family demographics.[7] Despite surgical and chemotherapeutic advances in treatment of pancreatic cancer, the prognosis has not greatly improved. There is an increasing number of new diagnoses, with only 20% of patients considered potentially curable.[8] When given a diagnosis of cancer, an individual's stress level may vary, as well as his or her ability to cope with that stress.[9] Individual patient stress may be related to fears of disease recurrences, cancer treatments, changes in family (dynamics) and personal relationships, work or legal issues, medication costs, and health insurance coverage.[10] In providing patient-focused care, it is recommended that healthcare practitioners maintain a supportive approach to offering information to the pancreatic cancer patient that should involve emotional, social, practical and spiritual support, as well as resources relating to coping with stress.[10] Discussions with the patient should include explanation about the individuals' diagnosis, prognosis, familiar risks, potential treatment options, and the positive and negative outcomes associated with treatment.

Stress is a subjective and implicit phenomenon that has been reported in the literature and defined in various ways. Stress is a consequence[11] that results from the physical, social and psychological environmental demands that threaten an individual's ability to cope.[12,13] Psychological distress is reported to negatively impact psychological well-being,[14] and emotional distress[15,16] is described as unpleasant emotional experiences of a physical, social, psychological or spiritual nature that may also interfere with an individual's ability to cope with cancer symptoms and treatment therapies.[17] Symptoms related to psychological distress are characterized by emotional suffering that can lead to depression and anxiety.[18] Despite the various definitions of stress,[19,20] consistent themes may be identified. In this qualitative systematic review, psychological stress will be defined as the experiences of unpleasant, emotional or environmental demands related to age as well as physical, social, psychological or spiritual occurrences that are reported by patients with pancreatic cancer during the pre-and post-diagnosis interventions and follow-up care.

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Patients with pancreatic cancer are commonly known as a high-risk group requiring psychological support services, but there is limited evidence quantifying or describing the psychological stress they experience. In several qualitative studies, these patients have described their stress as a separate symptom,[21] as emanating from the anticipation of treatments,[22] due to their physical symptoms,[23,24] as well as concerns about trust[25] and healthcare decision-making.[26] Patients also reported stress from situations involving caregiving and family issues[27] and from personal feelings about their anticipated death.[28,29] However, findings reported in related studies of pancreatic cancer patients show an increased risk of depression and anxiety,[20,29] suicide,[30] as well as significantly higher levels of depression compared to other advanced gastrointestinal, colonic and other abdominal tumors.[29,31,32]

There is growing evidence suggesting that a relationship exists between psychological stress and site-specific cancer mortality.[5,18] In a 19-year surveillance study, psychological stress experiences were reported by two-thirds of the sample (n = 4363) on one or more occasions during follow-up cancer care.[5] Similar studies found consistency between psychological stress and the development of long-term clinical depression and anxiety.[5,8] Stress negatively impacts the immune system and alters brain function in relation to the hypothalamic-pituitary-adrenocortical axis and the sympathetic nervous system. There is a release of glucocorticoid catecholamine and other factors that may impact malignancy.[9] Tumor activity increases the secretion of pro-inflammatory cytokines and Il-6 creating changes in the tryptophan-kynurenine, glutamate and serotonin pathways. This activity is related to various symptoms such as depression,[32,33] fatigue, lethargy, anorexia, impaired concentration, sleeplessness and increased pain sensitivity in cancer patients.[9]

There is also evidence supporting the increasing awareness that the psychosocial stress experienced by cancer patients may be related to negative outcomes of their treatment and prognosis.[9,10] The late detection and diagnosis of pancreatic cancers negatively impact a patient's choice of treatment options. This limited choice of options increases patients' morbidity; therefore, they may be ill or die before they can be included in studies identifying their stress experiences. Given the severity of pancreatic cancer, psychological support and stress management resources may improve the life quality and outcomes for this population of cancer patients. Identifying patients' experiences of stress would improve understanding of factors relating to their stress and enhance the oncology providers' ability to emotionally and psychologically address patients' needs. Providing appropriate education and social resources for these patients could improve the quality of their care. Communicating realistic expectations about potential treatments and outcomes could improve patients' ability to make decisions that could impact the remainder of their life.

A preliminary search of databases, including CINAHL, PubMed, the Cochrane Library and the *JBI Database of Systematic Reviews and Implementation Reports,* revealed that no systematic review has been conducted on the phenomena of interest over the last three years, or is in progress. There is limited literature in clinical gastroenterology and oncology practice that describes our phenomena of interest at pre-diagnosis, diagnosis, treatment, and post-treatment and follow-up care. For these reasons, this systematic review seeks to identify factors associated with psychological stress, reported by the pancreatic patient during pre-

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and post-diagnosis interventions and the post-treatment period defined as survivorship. The outcome of this review protocol will address this gap in the literature and provide insight into psychological stress of patients with pancreatic cancer at any time point in the continuum of their care.

Review questions

i. What are the experiences and perceptions of psychological stress reported by patients with pancreatic cancer?

ii. Are there differences in the experiences and perceptions of psychological stress reported by patients with pancreatic cancer at time points from pre-diagnosis, diagnosis, treatment, post-treatment and/or follow-up care and different demographics (i.e. age, sex)?

Inclusion criteria

Participants

This review will consider qualitative studies that include patients who have experienced stress as a result of pancreatic cancer. Studies will be considered regardless of a patient's age, sex or ethnicity.

Phenomena of interest

This review will consider studies that describe the stress experiences of patients with pancreatic cancer at time points from pre-diagnosis, diagnosis, treatment, post-treatment or follow-up care. Several definitions of stress exist.[6,19] This review defines psychological stress experiences as the unpleasant emotional or environmental demands related to physical, social, psychological or spiritual occurrences that are reported by cancer patients.

Context

Clinical settings may be varied and could include comprehensive cancer facilities, large metropolitan and general medical centers, or local primary care centers. The Herschblach *et al.*[34] study found significant differences in patient-reported psychological stress between treatment facilities. This may include varied methods[17,18] and assessments[20] to collect information associated with the documents of a patient's experiences of psychological stress or differences between general medical centers, outpatient clinics and palliative care centers.[33] This review will include studies that report screening initiatives to identify stress among patients with pancreatic cancer.[18]

Types of studies

This review will consider studies that focus on qualitative data including, but not limited to, designs such as phenomenology, grounded theory, ethnography, action research and feminist research.

In the absence of research studies, other texts included in gray literature, consisting of theses, policy, editorials, opinion papers and reports, expert opinion and progress notes, will

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be considered for review. All studies that are not written in English or do not have an English translation will be excluded from this review.

Methods

This review will follow JBI methodology for systematic reviews of qualitative evidence.[35]

Search strategy

Preliminary searches using keywords revealed limited studies in the time frame between 2007 and date of literature search. Therefore, all literature published and unpublished up to the date of the search will be considered for inclusion in this review.

The search strategy aims to find both published and unpublished qualitative studies and will also consider mixed method studies. However, only the qualitative data from mixed-methods studies will be used in the data analysis. A three-step search strategy will be utilized in this review. An initial limited search of MEDLINE and CINAHL will be undertaken followed by analysis of the text words contained in the title and abstract, and of the index terms used to describe each article. A second search using all identified keywords and index terms will then be undertaken across all included databases. Thirdly, the reference list of all identified reports and articles will be searched for additional studies. Studies published in all languages will be included, only if there is an English translation associated with the study. Studies published from the inception of each database searched will be considered for inclusion in this review (Appendix I).

Information sources

The databases to be searched for published sources include PubMed, CINAHL, Cochrane, Web of Science, Embase, Scopus, BioMed Central and PsycINFO. The databases to be searched for gray literature include: Biosis, OpenGrey, Open Access Theses and Dissertations, and WorldCat.

Study selection

Following the search, all identified citations will be collated and uploaded into EndNote X9 (Clarivate Analytics, PA, USA) and duplicates removed. Titles and abstracts will then be screened by two independent reviewers for assessment against the inclusion criteria for the review. Studies that may meet the inclusion criteria will be retrieved in full and their details imported into the JBI System for the Unified Management, Assessment and Review of Information (JBI SUMARI; JBI, Adelaide, Australia). The full text of selected studies will be retrieved and assessed in detail against the inclusion criteria by two independent reviewers. Full-text studies that do not meet the inclusion criteria will be excluded, and reasons for exclusion will be provided in an appendix in the final systematic review report. Included studies will undergo a process of critical appraisal. The results of the search will be reported in full in the final report and presented in a Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram.[36,37] Any disagreements that arise between the reviewers will be resolved through discussion or with a third reviewer.

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Assessment of methodological quality

Qualitative papers selected for retrieval will be assessed by two independent reviewers for methodological validity prior to inclusion in the review using the JBI Critical Appraisal Checklist for Qualitative Research.[35] Any disagreements that arise between the reviewers will be resolved through discussion or with a third reviewer. The results of critical appraisal will be reported in narrative form and in a summary table.

Data extraction

Qualitative data will be extracted from papers included in the review using the meta-aggregation approach.[37,38] Studies will include all types of stress described by pancreatic cancer patients, as well as specific details about the interventions, populations, study methods and outcomes of significance to the review question and specific objectives. Coding will be assigned to any differences in the experiences and perceptions of psychological stress reported by pancreatic patients at time points as follows: i) pre-diagnosis, ii) diagnosis, iii) pre- and post-interventions involving treatment, and iv) post-treatment and follow-up care. As findings and their illustrations are extracted, each study will be assigned a level of credibility using the JBI levels of credibility.[35,37,38] Authors will be contacted to request missing or additional data, where required.

Data synthesis

Qualitative research findings will, where possible, be pooled. This will involve the aggregation or synthesis of findings to generate a set of statements that represent that aggregation, through assembling the findings rated according to their quality, and categorizing these findings based on similarity in meaning. If the number of studies and study findings allow for it, separate syntheses will be constructed at the various time points in the following coded format: i) pre-diagnosis, ii) diagnosis, iii) pre- and post-interventions and iv) post-treatment and follow-up care. These categories are then subjected to a meta-synthesis to produce a single comprehensive set of synthesized findings that can be used as a basis for evidence-based practice. Where textual pooling is not possible, the findings will be presented in narrative form.

Assessing confidence in the findings

The final synthesized findings will be graded to establish confidence in the output of qualitative research synthesis and presented in a Summary of Findings. Included in the Summary of Findings will be the title, population, phenomena of interest and context for the specific review. Each synthesized finding from the review is then presented along with the type of research informing it, a score for dependability, credibility and the overall ConQual score.[38]

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Appendix I:: Search strategy for PubMed

((((((((((((((Pancreatic Neoplasm) OR Pancreas Neoplasms) OR Neoplasm, Pancreas) OR Neoplasms, Pancreas) OR Pancreas Neoplasm) OR Neoplasms, Pancreatic) OR Cancer of Pancreas) OR Pancreas Cancers) OR Pancreas Cancer) OR Cancer, Pancreas) OR Cancers, Pancreas) OR Pancreatic Cancer) OR Cancer, Pancreatic) OR Cancers, Pancreatic) OR Pancreatic Cancers) OR Cancer of the Pancreas) OR "Pancreatic Neoplasms"[Mesh])) AND ((((((((((((((Psychological Stresses) OR Stresses, Psychological) OR Life Stress) OR Life Stresses) OR Stress, Life) OR Stresses, Life) OR Stress, Psychologic) OR Psychologic Stress) OR Psychological Stress) OR Mental Suffering) OR Suffering, Mental) OR Suffering) OR Anguish) OR Emotional Stress) OR Stress, Emotional) OR "Stress, Psychological"[Mesh]))

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